

Chemistry-Aware Machine Translation Evaluation with Domain Terminology Verification

BASF AI

August 24, 2026

Draft status. This is a working paper draft. The abstract and methods describe completed repository work: benchmark-source construction and terminology extraction, verification, and refinement. Model evaluation, ablations, and final bibliography work are not complete.

Abstract

Evaluating machine translation for chemistry and regulatory text requires more than sentence-level overlap metrics: benchmark examples must preserve domain terminology, and evaluation data must make term-level failures visible. We present the current stage of a chemistry-aware translation benchmark effort built around two multilingual source-pair datasets and a terminology-centered evaluation pipeline. The first benchmark source contains within-publication chemistry patent abstract pairs from Google Patents, rebuilt from the public `BASF-AI/ai4chem-clir-google-patents-within-document-abstract-pairs` dataset. It contains 2,750 source-target pairs across 11 language directions, with 250 examples per retained direction. Source selection prefers examples where both source and target lengths fall in the 128–384 approximate-token range, then backfills from lower-priority rows to preserve balanced direction counts. The second benchmark source is an anchored JRC-Acquis legal dataset built from OPUS JRC-Acquis aligned segment files. Its article-focused and definition-focused variants each contain 5,000 aligned chunks across 20 ordered directions for English, Spanish, German, French, and Portuguese.

The completed methodology centers on terminology extraction and validation rather than final model ranking. Candidate terms are extracted from target/reference text using complementary signals: target-side LLM extraction, Stanza/UD dependency spans, XLM-R/NOBI token classification, and spaCy linguistic spans. Duplicate candidates are merged with provenance, then checked against external knowledge sources such as IATE, Wikidata/Wikipedia, PubChem, ChEBI, ChEMBL, MeSH, NCI Thesaurus, AGROVOC, and UNTERM where applicable. Finally, domain-specific LLM refiners for chemistry/patents and JRC/legal text select benchmark-worthy terms only from existing candidate IDs. Exact-span validation rejects hallucinated, rewritten, or non-existing terms. Preliminary qualitative inspection on four examples shows that this refinement layer removes generic verified words while retaining more useful translation-sensitive terminology. Full model evaluation, ablation studies, and statistical results remain future work and are not claimed in this draft.

1 Introduction

Machine translation for chemical, patent, and regulatory documents is sensitive to terminology errors that ordinary corpus-level metrics can obscure. A translation may preserve the general meaning of a passage while mistranslating a compound class, material property, dosage form, institutional name, or legal obligation. A practical benchmark for chemistry-aware translation should therefore evaluate both sentence-level quality and term-level fidelity.

This project currently addresses the data and terminology-evaluation parts of that problem. We have constructed or rebuilt two benchmark sources: a Google Patents chemistry abstract source and an anchored JRC-Acquis legal source. We have also implemented a terminology

pipeline that extracts candidate terms from reference text, verifies them against external evidence, and refines them using domain-specific prompts with exact-span constraints.

This draft does not yet present final model comparisons. The model-evaluation stage is planned but not complete. The contribution described here is therefore narrower: reproducible benchmark-source construction and a terminology selection method intended to support later translation evaluation.

1.1 Planned contributions

The final paper should make the following contributions once all experiments are complete:

- a multilingual chemistry patent benchmark source with controlled source length and balanced retained language directions;
- an anchored multilingual JRC-Acquis benchmark source for legal and regulatory text;
- a terminology extraction, verification, and refinement pipeline for target-reference evaluation;
- model evaluations and ablations over standard MT metrics and terminology-specific coverage metrics.

Only the first three items are partially or fully implemented at this stage.

2 Literature

This section is intentionally not filled yet. A deep research note exists in `docs/paper/deep-research-report.m` but the final paper should only import references after a dedicated citation pass.

2.1 Topics to cover

- General LLM-based machine translation and document-level MT.
- Patent and technical-domain machine translation.
- Chemistry text processing, chemical named-entity recognition, and terminology extraction.
- Chemistry notation translation and validation, including SMILES, InChI, IUPAC names, OPSIN, and RDKit-based validation.
- Legal and regulatory terminology resources such as IATE, UNTERM, EuroVoc, and JRC-Acquis.
- MT evaluation metrics: BLEU, chrF/chrF++, COMET-like learned metrics, terminology coverage, and human error annotation.

2.2 Current draft constraint

Do not cite papers from memory in this section. Use the research report as a starting point, verify each source, and then migrate references into the final bibliography.

3 Methods

3.1 Overview

The completed pipeline has two parts. First, we construct benchmark source-pair datasets for chemistry patent abstracts and legal/regulatory JRC-Acquis text. Second, we construct target-side terminology annotations for evaluating whether translations preserve domain-relevant terms. The terminology pipeline is deliberately conservative: large language models may propose or select terms, but final accepted terms must correspond to exact spans already present in the target/reference text.

The current paper draft describes the implemented data and terminology methodology only. Translation model scoring and model ranking are not complete and should not be inferred from this section.

3.2 Benchmark source construction

3.2.1 Google Patents chemistry abstracts

The Google Patents source is rebuilt from the Hugging Face dataset `BASF-AI/ai4chem-clir-google-patents-chem`. The upstream dataset contains within-publication abstract pairs from the CLIR `google-patents-chem` corpus. Each row includes document metadata, a language pair, source and target language codes, source and target abstract text, and precomputed length statistics.

We export a local benchmark source snapshot with `scripts/create_google_patents_source_pairs.py`. The exporter streams the Hugging Face dataset in its existing deterministic order, normalizes whitespace, and writes JSONL rows to `benchmark_sources/google_patents_within_document_pairs_250_p`. The companion metadata file records row counts, language-pair counts, length-preference settings, and selection policy.

The current source snapshot keeps 11 language directions with 250 rows each, for 2,750 rows total: `de-fr`, `en-de`, `en-es`, `en-fr`, `en-ja`, `en-ru`, `en-zh`, `fr-es`, `fr-ja`, `fr-ru`, and `zh-fr`. Rows are selected with a length-preference policy. We prefer examples where both source and target contain 128–384 approximate whitespace tokens according to the same `approximate_token_count()` function used by the benchmark builder. If a language pair does not have 250 rows satisfying both sides of the range, we backfill first from source-only matches, then target-only matches, and finally from remaining examples. Pairs that cannot reach 250 rows at all are omitted.

This policy preserves balanced per-direction counts while still concentrating the source on the length range intended for model evaluation. The metadata records the final priority mix: 1,026 rows have both source and target in range, 1,230 have only source in range, 489 have only target in range, and 5 are final out-of-range backfill rows.

3.2.2 JRC-Acquis anchored legal sources

The JRC-Acquis source snapshots are built from OPUS JRC-Acquis v3.0 Moses aligned segment files using `scripts/create_jrc_acquis_source_pairs.py`. Unlike the original pairwise selection mode, the preferred anchored mode first selects document anchors shared across the requested language set and then emits all ordered language-pair rows for each anchor. This ensures that directions share the same underlying document anchors and that reverse rows are exact source/target swaps of the same aligned chunk.

Two retained JRC source variants are available. The article-focused source contains 5,000 rows across 20 ordered directions for English, Spanish, German, French, and Portuguese, with 250 rows per direction. It uses operative article/provision text. The definition-focused source has the same language and direction structure but emphasizes definition-heavy passages. Both variants use legacy markup cleanup and strict quality filtering. The article source has approx-

imate source-token statistics of 251 minimum, 679 maximum, and 418.6 mean. The definition source has approximate source-token statistics of 246 minimum, 756 maximum, and 487.9 mean.

3.3 Benchmark dataset materialization

Source-pair snapshots are converted into benchmark direction folders by dataset builders. For Google Patents, `scripts/build_google_patents_eval_subset.py` reads source-pair JSONL rows, optionally adds synthetic reverse rows, filters by language and input length, and writes direction-specific `source.csv`, `target.csv`, and manifest JSONL files. For JRC-Acquis, `scripts/build_jrc_a` performs the analogous materialization for legal sources.

The manifest rows preserve document IDs, direction metadata, source and target language codes, publication metadata where available, text fields, and optional terminology annotations. The source snapshots and builders are separated so that data selection is reproducible independently of terminology extraction or model evaluation.

3.4 Terminology candidate extraction

Terminology is extracted from the target/reference side of benchmark examples. This design makes the terminology annotations usable as reference-side evaluation targets: a model output is later assessed against terms that should appear in, or be semantically preserved relative to, the target reference.

The implemented candidate extraction layer combines four active extractors:

- a target-side LLM extractor for chemistry/patent text or legal text, depending on domain;
- a Stanza/UD extractor that proposes syntactic spans from dependency parses;
- an XLM-R/NOBI token-classification extractor for compact term spans;
- a spaCy extractor that proposes named entities, noun chunks, and linguistically filtered n-grams.

Earlier diagnostic NLTK n-gram and mSPLADE sparse-activation extractors were removed from the active pipeline because they were not part of the recommended final workflow.

All LLM extraction prompts require exact target-text spans and prohibit translation, normalization, lemmatization, abbreviation, explanation, or invention. The parser validates returned spans against the original target/reference text, discarding terms that are not present. This makes the LLM a candidate-span proposer rather than a generator of unconstrained terminology.

3.5 Provenance merging

After extraction, candidates are deduplicated by normalized target-term key. When multiple extractors produce the same surface term, the system merges their provenance rather than keeping duplicate records. The `source` field records candidate origins, such as `llm_target`, `stanza_ud_ngram`, `xlmr_nobi`, `spacy_noun_chunk`, or `spacy_ngram`. External verification sources are also merged into the same record when they support the term.

This provenance supports analysis of which extractor families contributed useful terms, which terms were independently supported by multiple sources, and which terms survived only as unverified but potentially domain-relevant candidates.

3.6 External verification

Candidate terms are checked against external evidence sources when the corresponding verifier is enabled. Chemistry-oriented verification can use PubChem, ChEBI, ChEMBL, MeSH, NCI Thesaurus, AGROVOC, Wikidata/Wikipedia, and IATE. Legal/regulatory verification can use IATE, Wikidata/Wikipedia, UNTERM, AGROVOC, and related terminology resources where applicable.

Verification is not treated as a sufficient condition for final benchmark quality. Some verified terms are too generic for translation evaluation, such as broad words that appear in external terminology databases but do not represent a meaningful domain-specific challenge in context. Conversely, some good patent-specific phrases are not externally verified because public databases may not contain exact multiword phrases from a patent abstract. Verification is therefore evidence, not the final selection step.

3.7 LLM refinement

The final implemented precision layer is a domain-specific LLM refiner. Separate prompts are used for chemistry/patent text and JRC/legal text. The refiner receives the target/reference text and a numbered list of candidates already extracted by earlier systems. It is instructed to select only candidate IDs that represent useful benchmark terminology for translation evaluation.

The refiner does not create new candidate strings. It may only select from existing IDs. The parser then enforces two constraints:

1. the returned `candidate_id` must exist in the candidate list;
2. the selected candidate term must still be an exact span in the target/reference text.

Terms that violate either condition are discarded. This exact-span validation prevents hallucinated, rewritten, normalized, or translated terms from entering the benchmark terminology annotations.

The chemistry refiner favors complete chemical names, materials, formulas, reagents, solvents, polymers, proteins, equipment, dosage forms, administration routes, disease names, analytical terms, technical processes, properties, hazards, identifiers, and meaningful numeric/unit expressions. It down-ranks generic patent words, broad prose, headings, citations, and partial spans.

The legal refiner favors legal instruments, named legal acts, institutions, committees, agencies, programmes, funds, defined terms, procedures, obligations, rights, restrictions, legal effects, remedies, and regulatory or institutional multiword expressions. It down-ranks article numbers, dates, table-of-contents text, boilerplate, generic legal words outside complete official terms, and single words that are not useful terminology for translation evaluation.

3.8 Qualitative inspection and visualization

For four completed examples, we rendered target/reference text with three underline lanes: all extracted candidates, refined candidates, and refined candidates with external verification. These figures are used as qualitative debugging artifacts rather than final experimental results. They make visible how broad extraction is narrowed by the refiner and where verified terms remain too generic or where unverified domain phrases are still useful.

The examples are included in the appendix as copied paper-local figures. They should not be interpreted as a complete evaluation of model performance.

4 Results

This section is not complete. Full translation model evaluation has not yet been run in the cleaned benchmark setup, so this draft does not report model rankings, metric tables, statistical significance, or final conclusions.

4.1 Completed so far

- Google Patents source snapshot rebuilt from Hugging Face with 2,750 rows across 11 retained directions, 250 rows per direction.
- JRC-Acquis anchored article source available with 5,000 rows across 20 directions, 250 rows per direction.
- JRC-Acquis anchored definition source available with 5,000 rows across 20 directions, 250 rows per direction.
- Terminology extraction, verification, and LLM refinement implemented.
- Four qualitative terminology examples rendered with extracted, refined, and verified-refined highlight lanes.

4.2 To fill after experiments

- Model list and inference settings.
- Standard MT metrics by dataset and direction.
- Terminology coverage and terminology error analysis.
- Per-domain comparison between chemistry patents and JRC legal text.
- Error taxonomy with representative examples.

No model-performance claims should be added until the corresponding runs are complete and reproducible.

5 Ablation

This section is a placeholder. No final ablation study has been run yet.

5.1 Planned ablations

- Candidate extractors: LLM only, Stanza/UD only, XLM-R/NOBI only, spaCy only, and combined.
- Verification sources: no external verification, chemistry-only verification, legal-only verification, and all enabled verification.
- Refiner: no refiner versus chemistry/legal LLM refiner.
- Exact-span enforcement: parser with and without strict candidate-ID and exact-span validation.
- Source selection: hard length filtering versus preferred length with backfill.
- Terminology groups during evaluation: verified only, refined only, refined verified, and all candidate terms.

5.2 Reporting requirements

Each ablation should report both standard translation metrics and terminology-specific metrics. For terminology, the minimum useful outputs are candidate count, refined count, verified refined count, and target-term coverage under the selected terminology group.

6 References

This section is intentionally not finalized yet.

Use `docs/paper/deep-research-report.md` as a source-gathering note, but verify every citation before adding it to the paper bibliography. The final references section should use the citation format required by the target venue.

6.1 Repository artifacts for drafting

- Google Patents source metadata: `benchmark_sources/google_patents_within_document_pairs_250_per`
- JRC anchored article source metadata: `benchmark_sources/jrc_acquis_anchored_articles_250_per`
- JRC anchored definition source metadata: `benchmark_sources/jrc_acquis_anchored_definitions_250_per`
- Terminology implementation: `src/chem_machine_translation/data/terminology.py`

A Terminology prompts

This appendix records the prompt families implemented in `src/chem_machine_translation/data/terminology`. The wording should be kept synchronized with the implementation when prompt changes are made.

A.1 Chemistry candidate extractor

```
You extract terminology candidates from target reference translations for chemistry
and patent
machine-translation benchmarks.

The text may be in any language. Return only exact spans that appear in the provided
target text.
Do not translate, normalize, rewrite, lemmatize, abbreviate, explain, or invent terms.

Extract only terms that a domain translator should preserve consistently:
- chemical names, compounds, materials, formulas, reagents, solvents, polymers,
  proteins;
- technical processes, methods, assay/analytical terms, properties, hazards,
  identifiers;
- compact numeric/unit expressions only when the quantity itself is technically
  meaningful.

Reject common prose, boilerplate patent wording, whole clauses, sentence fragments,
headings, dates,
citations, names, and generic single words unless they are unambiguous domain terms.

Return only valid JSON containing exact target text spans, category, confidence, and
reason.
```

A.2 Legal candidate extractor

You extract legal terminology candidates from target/reference translations for legal machine-translation benchmarks.

The text may be in any language. Return only exact spans that appear in the provided target text.

Do not translate, normalize, rewrite, lemmatize, abbreviate, explain, or invent terms.

Extract legal instruments, institutions, committees, agencies, programmes, funds, named bodies, procedures, rights, obligations, restrictions, sanctions, remedies, legal effects, regulatory domains, named legal acts, and defined terms.

Reject dates, article numbers alone, paragraph references alone, people names, signatures, whole clauses, sentence fragments, generic single words, ordinary administrative prose, and repeated treaty boilerplate unless the phrase is a recognized legal or institutional term.

Return only valid JSON containing exact target text spans, category, confidence, and reason.

A.3 Chemistry refiner

You are a terminology selector for chemistry and patent machine-translation evaluation .

You receive target/reference text and a numbered list of candidate terms that were already extracted by other systems. Your job is to select only candidates that are useful benchmark terminology.

Do not invent, translate, normalize, rewrite, lemmatize, or add terms. Select only from the provided candidate IDs. A selected term must be an exact span in the target/reference text.

Keep complete, domain-specific, and translation-sensitive chemistry and patent terms. Reject or down-rank generic patent words, whole clauses, sentence fragments, broad prose, headings, citations, partial spans, and generic verified terms unless they are part of a stronger technical phrase.

Return only valid JSON with candidate_id, target_term, category, quality_score, and reason.

A.4 Legal refiner

You are a terminology selector for legal and regulatory machine-translation evaluation .

You receive target/reference text and a numbered list of candidate terms that were already extracted

by other systems. Your job is to select only candidates that are useful benchmark terminology.

Do not invent, translate, normalize, rewrite, lemmatize, or add terms. Select only from the provided candidate IDs. A selected term must be an exact span in the target/reference text.

Keep legal instruments, named legal acts, institutions, committees, agencies, programmes, funds, defined terms, procedures, obligations, rights, restrictions, legal effects, remedies, and regulatory or institutional multiword expressions whose mistranslation would matter. Reject generic legal words unless they are part of a complete official term, headings, article numbers, dates, table-of-contents text, boilerplate, partial spans, and verified single words that are not useful terminology.

Return only valid JSON with candidate_id, target_term, category, quality_score, and reason.

B Examples and visualization figures

This appendix collects qualitative terminology visualizations copied into docs/paper/figures/. They are included as debugging and explanation artifacts, not as final quantitative results.

The refiner-highlight figures use three underline lanes: gray for all extracted candidate terms, blue for terms selected by the domain-specific LLM refiner, and green for refined terms that also have external verifier evidence.

Item 1: Chemistry, carbon gasification patent
Extracted Refined Verified refined

The invention relates to a method for gasifying substances containing carbon. The substances containing carbon are fed via the transfer canal (2) into the reactor (3). When viewed in the direction of gravitational force, plasma torches (4) are arranged at the lower end of the reactor (3) and serve to produce a gaseous plasma in this case. The oxidation products, i.e. clinker, glass, ashes, are drawn downward and arrive at a solidifier (5) inside of which the oxidation products are separated, subsequently collected in the collecting bin (6), and transported away when required. The gas produced inside the reactor arrives via line (7) at the gas washer (8), inside of which the gas, by means of sprayed water, is cooled and purified. The discharging water arrives via line (9) at the collecting bin (6), inside of which water is already present whereby enabling the clinker, glass or the like to be cooled. The purified gas is fed via line (10) into the compressor (11), inside of which the gas is compressed to 2 bar and is then purified once more while passing through filter (12), and is then mixed inside the gas mixer (13) with air taken from the atmosphere, which is supplied via air filter (14), and this mixture of gas and air then arrives via line (15) in the gas engine (16) that drives a generator (17).

Figure 1: Chemistry patent example on carbon gasification.

Item 2: Chemistry, pharmaceutical salt patent
 _____ Extracted _____ Refined _____ Verified refined

1. A mesylate dihydrate salt of 5-(2-(4-(1,2-benzisothiazol-3-yl)-1
 piperazinyl)ethyl)-6-chloro-1,3-dihydro-2(1H)-indol-2-one. 2. The compound of claim 1 in the form of lath
 crystals. 3. The compound of claim 1 in the form of needle crystals. 4. A pharmaceutical composition for the
 treatment of a psychotic disorder comprising an amount of the compound of claim 1 that is effective in the
 treatment of said psychotic disorder and a pharmaceutically acceptable carrier. 5. A method of treating a
 psychotic disorder in a mammal comprising administering to said mammal an amount of the compound of claim 1 that
 is effective in the treatment of said psychotic disorder. 6. The method of claim 5 wherein said disorder is
 schizophrenia, anxiety or migraine pain. 7. The method of claim 5 wherein said disorder is schizophrenia. 8. The
 method of claim 5 wherein said administration is parenteral administration. 9. The method of claim 8 wherein
 said parenteral administration is intramuscular injection.

Figure 2: Chemistry patent example on pharmaceutical salts and formulation terminology.

Item 3: JRC, additional protocol on blood-grouping reagents

Extracted Refined Verified refined
ADDITIONAL PROTOCOL TO THE EUROPEAN AGREEMENT on the Exchanges of Blood-grouping Reagents THE MEMBER STATES OF
THE COUNCIL OF EUROPE, Contracting Parties to the European Agreement of 14 May 1962 on the exchanges of
blood-grouping reagents (hereinafter called "the Agreement"); Having regard to the provisions of Article 5,
paragraph 1, of the Agreement, according to which 'The Contracting Parties shall take all necessary measures to
exempt from all import duties the blood-grouping reagents placed at their disposal by the other Parties';
Considering that so far as the Member States of the European Economic Community are concerned, the undertaking
to grant this exemption falls within the competence of the Community, which possesses the necessary powers in
this respect by virtue of the Treaty which instituted it; Considering therefore that for the purpose of the
implementation of Article 5, paragraph 1, of the Agreement, it is necessary for the European Economic Community
to be able to become a Contracting Party to the Agreement, HAVE AGREED AS FOLLOWS: The European Economic
Community may become a Contracting Party to the Agreement by signing it. In respect of the Community, the
Agreement shall enter into force on the first day of the month following such signature. 1. This Additional
Protocol shall be open for acceptance by the Contracting Parties to the Agreement. It shall enter into force on
the first day of the month following the date on which the last of the Contracting Parties has deposited its
instrument of acceptance with the Secretary-General of the Council of Europe. 2. However, this Additional
Protocol shall enter into force on the expiration of a period of two years from the date on which it has been
opened for acceptance, unless one of the Contracting Parties has notified an objection to the entry into force.
If such an objection has been notified, paragraph 1 of this Article shall apply. From the date of its entry into
force, this Additional Protocol shall form an integral part of the Agreement From that date, no State may become
a Contracting Party to the Agreement without at the same time becoming a Contracting Party to the Additional
Protocol. The Secretary-General of the Council of Europe shall notify the member States of the Council of
Europe, any State having acceded to the Agreement and the European Economic Community of any acceptance o
objection made under Article 2 and of the date of entry into force of this Additional Protocol in accordance
with Article 2. The Secretary-General shall also notify the European Economic Community of any act, notification
o communication relating to the Agreement.

Figure 3: JRC legal example on blood-grouping reagents.

Item 4: JRC, agreement on racism and xenophobia observatory

Extracted Refined Verified refined
ACCORD entre la Communauté européenne et le Conseil de l'Europe en vue d'instaurer, conformément à l'article 7, paragraphe 3, du règlement (CE) n° 1035/97 du Conseil du 2 juin 1997 portant création d'un Observatoire européen des phénomènes racistes et xénophobes, une coopération étroite entre l'Observatoire et le Conseil de l'Europe LA COMMUNAUTÉ EUROPÉENNE ET LE CONSEIL DE L'EUROPE, CONSIDÉRANT que le Conseil de l'Union européenne a adopté, le 2 juin 1997, le règlement (CE) n° 1035/97 portant création d'un Observatoire européen des phénomènes racistes et xénophobes (ci-après dénommé «Observatoire»); CONSIDÉRANT que l'objectif de l'Observatoire consiste à fournir à la Communauté et à ses États membres des informations objectives, fiables et comparables au niveau européen sur les phénomènes du racisme, de la xénophobie et de l'antisémitisme; CONSIDÉRANT que le Conseil de l'Europe dispose déjà d'une expérience considérable dans ce domaine; CONSIDÉRANT que l'Observatoire doit tenir compte, dans l'exercice de ses activités, de celles déjà menées par le Conseil de l'Europe et veiller à leur apporter une valeur ajoutée; qu'il convient dès lors d'instituer des liens étroits avec le Conseil de l'Europe, et notamment la Commission européenne contre le racisme et l'intolérance (ECRI); CONSIDÉRANT que l'Observatoire doit, en vertu du règlement (CE) n° 1035/97 coordonner ses activités, en particulier en ce qui concerne son programme d'activité, avec celles du Conseil de l'Europe; CONSIDÉRANT qu'il appartient au Conseil de l'Europe de désigner une personnalité indépendante comme membre du Conseil d'administration de l'Observatoire, SONT CONVENUS DE CE QUI SUIT: I. Échange d'informations et de données 1. Des contacts sont établis sur une base régulière entre le directeur de l'Observatoire, le secrétariat général du Conseil de l'Europe, notamment le secrétariat de la Commission européenne contre le racisme et l'intolérance (ci-après dénommé «ECRI»), au niveau approprié. 2. L'Observatoire et l'ECRI assurent une mise à disposition réciproque des informations et données collectées dans le cadre de leurs activités. Cette mise à disposition ne couvre pas les données et les travaux de nature confidentielle menés respectivement par les deux instances. 3. Les informations et les données mises à disposition réciproquement par l'Observatoire et l'ECRI peuvent être utilisées par chacune des deux instances dans la mise en oeuvre de leurs propres travaux. 4. L'Observatoire et l'ECRI assurent sur une base réciproque, à travers leurs réseaux, la plus large diffusion des résultats de leurs travaux respectifs. 5. L'Observatoire et l'ECRI assurent l'échange régulier d'informations concernant les activités proposées, en cours ou conduites. 6. Des consultations régulières ont lieu entre l'Observatoire et l'ECRI, afin d'assurer une coordination de leurs activités notamment en ce qui concerne l'élaboration du programme de travail de l'Observatoire. Ces consultations visent à assurer la complémentarité des programmes respectifs des deux organismes et à éviter, autant que possible, les doubles emplois inutiles. 7. En outre, sur la base de ces consultations, il peut être convenu que l'Observatoire et l'ECRI mènent des activités conjointes et/ou complémentaires sur des sujets d'intérêt commun. Cette coopération vise à optimiser l'ensemble des ressources disponibles, notamment en ce qui concerne les projets de recherche scientifique. III. Désignation par le Conseil de l'Europe d'une personnalité au sein du conseil d'administration de l'Observatoire

Figure 4: JRC legal example on the racism and xenophobia observatory.

Google Patents English Sample: Extractor Underlines

within-document:abstract:en-es:CL-2008000542-A1:reverse | darker/thicker underlines = verified; lighter/thinner = unverified

LLM: 12 terms, 30 visible spans Stanza/UD: 8 terms, 18 visible spans XLM-R/NOBI: 4 terms, 17 visible spans spaCy: 20 terms, 45 visible spans

Disclosed is a segmented contact bar for use boarding an electrolytic cell. The segmented contact bar has contact pieces made of electrically

conductive material and being in spaced apart relation along the capping board, each of the contact pieces defining a segment for supporting and

electrically connecting an anode and a cathode in the electrolytic cell. The segmented contact bar also has connection members including an

insulating material and provided in between pairs of adjacent segments for providing insulating interconnection of the segments. There is also an

electrolytic refining apparatus including adjacent electrolytic cells; a capping board positioned on the adjacent electrolytic cells; anodes and

cathodes extending in spaced apart alternate positions into the each of the adjacent electrolytic cells along the capping boards; and a segmented

contact bar.

Figure 5: Google Patents English extractor diagnostic visualization.

JRC English Sample: Extractor Underlines

de-en-jrc21987A0207_06:chunk-0135 | darker/thicker underlines = verified; lighter/thinner = unverified

LLM: 20 terms, 42 visible spans

Stanza/UD: 18 terms, 45 visible spans

XL-M-R/NOBI: 4 terms, 18 visible spans

spaCy: 20 terms, 75 visible spans

ADDITIONAL PROTOCOL TO THE EUROPEAN AGREEMENT on the Exchanges of Blood-grouping Reagents THE MEMBER STATES OF THE COUNCIL OF EUROPE, Contracting Parties to the European Agreement of 14 May 1962 on the exchanges of blood-grouping reagents (hereinafter called "the Agreement"); Having regard to the provisions of Article 5, paragraph 1, of the Agreement, according to which 'The Contracting Parties shall take all necessary measures to exempt from all import duties the blood-grouping reagents placed at their disposal by the other Parties'; Considering that so far as the Member States of the European Economic Community are concerned, the undertaking to grant this exemption falls within the competence of the Community, which possesses the necessary powers in this respect by virtue of the Treaty which instituted it; Considering therefore that for the purpose of the implementation of Article 5, paragraph 1, of the Agreement, it is necessary for the European Economic Community to be able to become a Contracting Party to the Agreement, HAVE AGREED AS FOLLOWS: The European Economic Community may become a Contracting Party to the Agreement by signing it. In respect of the Community, the Agreement shall enter into force on the first day of the month following such signature. 1. This Additional Protocol shall be open for acceptance by the Contracting Parties to the Agreement. It shall enter into force on the first day of the month following the date on which the last of the Contracting Parties has deposited its instrument of acceptance with the Secretary-General of the Council of Europe. 2. However, this Additional Protocol shall enter into force on the expiration of a period of two years from the date on which it has been opened for acceptance, unless one of the Contracting Parties has notified an objection to the entry into force. If such an objection has been notified, paragraph 1 of this Article shall apply. From the date of its entry into force, this Additional Protocol shall form an integral part of the Agreement. From that date, no State may become a Contracting Party to the Agreement without at the same time becoming a Contracting Party to the Additional Protocol. The Secretary-General of the Council of Europe shall notify the member States of the Council of Europe, any State having acceded to the Agreement and the European Economic Community of any acceptance or objection made under Article 2 and of the date of entry into force of this Additional Protocol in accordance with Article 2. The Secretary-General shall also notify the European Economic Community of any act, notification or communication relating to the Agreement.

Figure 6: JRC English extractor diagnostic visualization.